

# CHEYANNE SHARIAT

[cshariat@caltech.edu](mailto:cshariat@caltech.edu) ♦ [cheyanneshariat.github.io](https://cheyanneshariat.github.io)

## EDUCATION

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|-----------------------------------------------------------------|-------------|
| <b>Ph.D., Astrophysics</b> , California Institute of Technology | 2024 -      |
| <b>M.S., Astrophysics</b> , California Institute of Technology  | 2024 - 2026 |
| DOE Computational Science Graduate Fellow                       |             |
| Advisor: Kareem El-Badry                                        |             |
| <b>B.S., Physics</b> , University of California, Los Angeles    | 2021 - 2024 |
| <i>summa cum laude</i>                                          |             |
| Tisch Academic Excellence Scholar                               |             |
| Advisor: Smadar Naoz                                            |             |

## RESEARCH POSITIONS

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|---------------------------------------------|------------------------|
| <b>Graduate Student Researcher</b>          | 2024 -                 |
| California Institute of Technology          | <i>Pasadena, CA</i>    |
| <b>Undergraduate Student Researcher</b>     | 2021 - 2024            |
| University of California, Los Angeles       | <i>Los Angeles, CA</i> |
| <b>Summer Undergraduate Research Fellow</b> | 2023                   |
| NASA Jet Propulsion Laboratory (JPL)        | <i>Pasadena, CA</i>    |
| <b>Summer Undergraduate Research Fellow</b> | 2022                   |
| Caltech High Energy Astrophysics Group      | <i>Pasadena, CA</i>    |
| <b>High School Researcher</b>               | 2019 - 2022            |
| UCLA Neurology and Neurogenetics, Fogel Lab | <i>Los Angeles, CA</i> |

## HONORS & AWARDS

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|--------------------------------------------------------------------------|-------------|
| DOE Computational Science Graduate Fellowship                            | 2025 – 2029 |
| NSF Graduate Research Fellowship (offer declined)                        | 2025        |
| Raynor L. Duncombe Student Research Prize, AAS DDA                       | 2025        |
| Joshua and Beth Friedman Fellowship, Caltech                             | 2025        |
| Dean's Prize for Excellence in Research, UCLA                            | 2024        |
| Lindau Nobel Laureate Meetings Fellow, Lindau, Germany                   | 2024        |
| Helen Quinn Award for Undergraduate Research Theory (1st), APS           | 2023        |
| Summer Undergraduate Research Fellowship, NASA Jet Propulsion Laboratory | 2023        |
| Undergraduate Research Fellowship, UCLA                                  | 2022 – 2023 |
| Summer Undergraduate Research Fellowship, Caltech                        | 2022        |
| Steve Tisch Academic Excellence Scholarship, UCLA                        | 2021 – 2024 |

## GRANTS

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|----------------------------------------------------------|-------------|
| <b>Roman General Investigator Grant</b> , NASA/STScI, PI | 2027 – 2030 |
| <i>Binary Star Demographics in the Galactic Bulge</i>    |             |
| <b>Roman General Investigator Grant</b> , NASA/STScI, PI | 2027 – 2030 |
| <i>RACS: Roman Astrometric Companion Survey</i>          |             |

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|--------------------------------------------------------------------------------------|-------------|
| <b>Roman General Investigator Grant</b> , NASA/STScI, Co-I                           | 2027 – 2030 |
| <i>RAGHAB: Roman Analysis for a Galactic Hunt for Astrometric Binaries</i>           |             |
| <b>JWST General Observer Grant</b> , NASA/STScI, PI                                  | 2026 – 2029 |
| <i>Imaging and spectroscopy of a hypervelocity globular cluster ejected from M87</i> |             |

TALKS

|                                                                  |      |
|------------------------------------------------------------------|------|
| Stellar Thinkshop, Carnegie Observatories, Pasadena, CA          | 2026 |
| Astro Coffee, IAS, Princeton, NJ                                 | 2026 |
| Thursday Lunch Talk, Princeton, Princeton, NJ                    | 2026 |
| Stellar-Mass Black Holes Workshop, Talk, KITP                    | 2025 |
| The Lifecycle of Stellar Black Holes Conference, Talk, KITP      | 2025 |
| Carnegie ArXiv Tea, Pasadena, CA                                 | 2025 |
| Keck Science Meeting, Poster, UCLA                               | 2025 |
| Binary Stars in the Space Era, Keele, United Kingdom             | 2025 |
| Vasto Accretion Meeting, Vasto, Italy                            | 2025 |
| AAS Division on Dynamical Astronomy Meeting 56, Atlanta, GA      | 2025 |
| Undergraduate Research Week, UCLA                                | 2024 |
| Exoplanet Monthly Meeting, Northwestern, Evanston, IL            | 2024 |
| APS Conference for Undergraduate Women in Physics, San Diego, CA | 2024 |
| APS Far West Section, UCSD (Helen Quinn Award, 1st)              | 2023 |
| Division 3262 Seminar, NASA/JPL-Caltech                          | 2023 |
| Undergraduate Research Week, UCLA                                | 2023 |
| SURF Final Presentation, Caltech                                 | 2022 |
| HEAG Presentation Week, Caltech                                  | 2022 |

AWARDED TELESCOPE TIME

|                              |                               |
|------------------------------|-------------------------------|
| James Webb Space Telescope   | 24.8 hours (PI)               |
| Palomar Hale 200 inch, NGPS  | 8 nights (PI) 5 nights (Co-I) |
| Palomar Hale 200 inch, PHARO | 9 nights (PI)                 |
| Keck HIRESb                  | 9 nights (Co-I)               |
| Keck KCWI                    | 1 night (Co-I)                |

OBSERVING EXPERIENCE

|                                          |       |
|------------------------------------------|-------|
| Palomar Hale Telescope, PHARO – 8 nights | 2024- |
| Keck, HIRES – 2 nights                   | 2025- |

MENTORING

|                                |       |
|--------------------------------|-------|
| Joe Smith (Caltech Undergrad)  | 2025- |
| Noah East (UCLA Undergrad)     | 2025- |
| Omri Ratzkoff (UCLA Undergrad) | 2025- |

## JOURNAL REFEREE

ApJ, ApJL, MNRAS, A&A, A&AL (14 papers) 2023 -

## COLLABORATIONS

Sloan Digital Sky Survey 2026 -

LISA Consortium 2025 -

Euclid Consortium 2025 -

## TEACHING & OUTREACH

Graduate Teaching Assistant (Caltech) 2025

Ay 101: Physics of Stellar Interiors and Atmospheres

Graduate Peer Mentor (Caltech) 2025 -

LAUSD Guest Speaker 2023 -

James Monroe High School

Shenendoah Early Education Center

Stargazing Lectures (Caltech) 2023 -

Q&A Panelist

Telescope Operator

Diversity, Equity, and Inclusion (DEI) Committee 2023 - 2024

Undergraduate Representative, UCLA Physics and Astronomy Dept.

Los Angeles, CA

Undergraduate Learning Assistant 2022

Physics 1C, Electrodynamics, Optics, and Special Relativity, UCLA

Los Angeles, CA

## PUBLICATIONS (19 TOTAL; 12 FIRST-AUTHOR)

19. Ding, K., Gennaro, M., Avila, R. J., Ricotti, M., Beaton, R. L., Boyer, M. L., Brown, T. M., Calamida, A., Cassisi, S., Chandra, V., Cohen, R. E., Correnti, M., Crnojević, D., El-Badry, K., Geha, M., Guhathakurta, P., Kallivayalil, N., Kirby, E. N., McQuinn, K. B. W., Savino, A., **Shariat, C.**, Simon, J. D., and Weisz, D. R., “Probing the IMF in the Early Universe – Direct measurements in the Boötes I UFD with JWST/NIRCam”, *The Astrophysical Journal*, *submitted*, arXiv:2605.15069, 2026.
18. **Shariat, C.**, El-Badry, K., MacLeod, M., and Leiner, E., “Gaia astrometry disfavors a binary origin for long secondary periods”, arXiv:2604.09767, *Publications of the Astronomical Society of the Pacific*, 2026.
17. **Shariat, C.**, “OverCite: Add citations in LaTeX without leaving the editor”, arXiv:2604.15366, *Research Notes of the American Astronomical Society*, 2026.
16. Bhattacharjee, S., El-Badry, K., Fuller, J., **Shariat, C.**, Yamaguchi, N., “Thermally inflated accretors in post-mass transfer binaries: Abell 35 and its class revisited”, arXiv:2603.23756, *Publications of the Astronomical Society of the Pacific*, 2026.
15. **Shariat, C.** and El-Badry, K., “A global view of post-interaction white dwarf-main sequence binaries”, *Publications of the Astronomical Society of the Pacific*, vol. 138, no. 3, Art. no. 034202, 2026.
14. **Shariat, C.**, Ye, C. S., Naoz, S., and Rose, S., “Fast Radio Bursts from White Dwarf Binary Mergers: Isolated and Triple-Induced Channels”, *The Astrophysical Journal Letters*, vol. 1000, no. 1, Art. no. L17, 2026.
13. **Shariat, C.**, El-Badry, K., Bhattacharjee, S., “How precisely can we measure the ages of subgiant and giant stars?”, arXiv:2510.08675, *The Open Journal of Astrophysics*, 2026.
12. Holzknecht, L., Naoz, S., **Shariat, C.**, “Dynamical Pathways to the Misalignment of the VHS 1256-1257 System”, *The Astrophysical Journal*, vol. 998, no. 1, Art. no. 122, 2026.

11. **Shariat, C.**, El-Badry, K., Gennaro, M., Ding, K., Simon, J.D., Avila, R.J., Calamida, A., Cassisi, S., Correnti, M., Weisz, D.R., Geha, M., Kirby, E.N., Brown, T.M., Ricotti, M., McQuinn, K.B.W., Kallivayalil, N., Gilbert, K., Pacifici, C., Guhathakurta, P., Crnojević, D., Boyer, M.L., Beaton, R.L., Chandra, V., Cohen, R.E., Renzini, A., Savino, A., Tollerud, E.J., “Wide binaries in an ultra-faint dwarf galaxy: discovery, population modeling, and a nail in the coffin of primordial black hole dark matter”, *Publications of the Astronomical Society of the Pacific*, vol. 137, no. 10, Art. no. 104103, 2025.
10. Xuan, Z., **Shariat, C.**, Naoz, S., “From Wide Triples to UCXBs: Multimessenger Detection of Dynamically-formed Black Hole-White Dwarf Systems with LISA”, *The Astrophysical Journal*, vol. 995, no. 1, Art. no. 27, 2025.
9. **Shariat, C.**, El-Badry, K., Naoz, S., “10,000 Resolved Triples from Gaia: Empirical Constraints on Triple Star Populations”, *Publications of the Astronomical Society of the Pacific*, vol. 137, no. 9, Art. no. 094201, 2025.
8. **Shariat, C.**, El-Badry, K., Naoz, S., Rodriguez, A.C., van Roestel, J., “Cataclysmic Variables in Triples: Formation Models and New Discoveries”, *Publications of the Astronomical Society of the Pacific*, vol. 137, no. 7, Art. no. 074201, 2025.
7. Matsuno, T., Kemp, A., Tanikawa, A., **Shariat, C.**, El-Badry, K., Dodd, E., Helmi, A., Koch-Hansen, A.J., Yamaguchi, N., Yan, H., “Unevolved Li-rich stars at low metallicity: a possible formation pathway through novae”, *Astronomy and Astrophysics*, vol. 699, Art. no. A171, EDP, 2025.
6. **Shariat, C.**, Naoz, S., El-Badry, K., Akira Rocha, K., Kalogera, V., Stephan, A.P., Burdge, K., Angelo, I., “Triple Evolution Pathways to Black Hole Low-Mass X-ray Binaries: Insights from V404 Cygni”, *The Astrophysical Journal*, vol. 983, no. 2, 2025.
5. **Shariat C.**, Naoz, S., El-Badry, K., Rodriguez, A.C., Hansen, B. M. S., Angelo, I., Stephan, A. P., “Once a Triple, Not Always a Triple: The Evolution of Hierarchical Triples that Yield Merged Inner Binaries,” *The Astrophysical Journal*, vol. 978, no. 1, 2025.
4. Stephan, A. P., Martin, D. V., Naoz, S., Hughes, N. R., and **Shariat, C.**, “Two Novel Hot Jupiter Formation Pathways: How White Dwarf Kicks Shape the Hot Jupiter Population”, *The Astrophysical Journal Letters*, 2024
3. **Shariat C.**, Hasegawa, Y., Yu, T. Y. M., Hansen, B. M. S., Hu, R., “Predicting the Dominant Formation Mechanism of Multi-Planetary Systems,” *The Astrophysical Journal Letters*, vol. 964, no. 1, 2024.
2. **Shariat, C.**, Naoz, S., Hansen, B. M. S., Angelo, I., Michaely, E., Stephan, A. P., “Dynamical Evolution of White Dwarfs in Triples in the Era of Gaia,” *The Astrophysical Journal Letters*, vol. 955, no. 1, 2023.
1. Lazzarini, M., Hinton, K., **Shariat, C.**, Williams, B.F., Garofali, K., Dalcanton, J.J., Durbin, M., Antoniou, V., Binder, B., Eracleous, M., Vulic, N., Yang, J., Wik, D., Gasca, A., Kuauhtzin, Q., “Multiwavelength Characterization of the High-mass X-Ray Binary Population of M33”, *The Astrophysical Journal*, vol. 952, no. 2, 2023.